

Library Manifest

The purpose of this document is to list the libraries that we use.

Note: You only need to run: `npm install` for these libraries to install.

See: WIN2025 Installation Outline for detailed instructions.

Project Setup

- Use VS Code Editor
- Must Install NodeJS (Latest version 23.6.1)
 - To check version: `node -v`
- Clone from repository
- Make new branch

Vite Project

- Vanilla JavaScript
- React (Component-based architecture, Virtual DOM)
 - Already installed: `npm create vite@latest`
 - Already done: `cd projectdirectory`
 - Already installed: `npm install`
- Note: To update all dependencies: `npm update`

Hosting

- To run the development server, from the “main” directory:
 - To run the server: `npm run dev`

Additional Libraries

Three.js

- 3D web browser graphics (façade to WebGL)
 - Already installed: `npm install three`

React Three Fiber (RF3)

- Use React syntax for 3D scenes (High code reusability)
 - Already installed: `npm install @react-three/fiber`

Drei

- Drei adds additional helper functions to support RF3 (improved camera controls)
 - **Already installed:** `npm install @react-three/drei`

Cannon.js

- Physics engine (collision, gravity)
 - **Already installed:** `npm install cannon-es`

react-spring

- Support for smooth animations using physics
 - **Already installed:** `npm install @react-spring/web`

react-router-dom

- Adds the useNavigate hook for page navigation
 - **Already installed:** `npm install react-router-dom`

Tailwind CSS

- A utility-first CSS framework that allows you to design directly in your HTML/JSX with low-level utility classes. It's highly customizable and great for rapid prototyping.
 - **Already installed:** `npm install -D tailwindcss postcss autoprefixer`
- Add a `tailwind.config.js` file

```
/** @type {import('tailwindcss').Config} */
export default {
  content: [
    "./index.html",
    "./src/**/*.{js,ts,jsx,tsx}",
  ],
  theme: {
    extend: {},
  },
  plugins: [],
}
```

- Add Tailwind Directives to your `src/index.css` file
 - `@tailwind base;`
 - `@tailwind components;`
 - `@tailwind utilities;`
- Add to `package.json`

```
{
  "scripts": {
    "build:css": "npx tailwindcss -i ./src/styles.css -o ./dist/output.css --watch"
  }
}
```

- Build
 - Already installed: `npm run build:css`

Testing

React Testing for Beginners: Start Here!

<https://www.youtube.com/watch?v=8Xwq35cPwYg&t=2139s>

Vitest

- Framework that Supports ESM(JavaScript), TypeScript, and JSX (React)
 - **Already installed:** `npm i -D vitest`
 - **Already done:** Edit the `package.json`
 - Add to `"scripts": { }` section
 - `"test": "vitest",`
 - `"test:ui": "vitest -ui"`
 - **Already installed:** `npm i -D @vitest/ui`
- Note:
 - To run CLI tests: `npm run test`
 - To run UI tests: `npm run test:ui`

React Testing Library

- For testing react components
 - **Already installed:** `npm i -D @testing-library/react@14.2.0`

jsdom

- jsdom (DOM emulator)
 - **Already installed:** `npm i -D jsdom@24.0.0`
 - **Already done:** Add file `vitest.config.ts` containing below information

```
import { defineConfig } from 'vitest/config';

export default defineConfig({
  test: {
    environment: 'jsdom'
  }
})
```

jest-dom

- Improves test readability and simplifies DOM testing
 - **Already installed:** `npm i -D @testing-library/jest-dom`

Recommended YouTube Channels

Rabbit Hole Syndrome

Wael Yasmina

The issue occurring due to new React 19 version, because React 19 is not officially supporting @react-three/fiber yet.

To Resolve this

Downgrade to React 18

```
npm uninstall react react-dom
```

```
npm install react@18 react-dom@18
```

?18.2.0

Then install @react-three/fiber

For @react-three/drei after installing it you might get an error saying three-mesh-bvh@0.7.8 is deprecated, just simply use v0.8.0 version

```
npm install three-mesh-bvh@0.8.0
```

